Part 5. Interaction Planning

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Principles

Interaction lets Natalia choose what to see, but every action has a cognitive cost (course 07). We added only interactions that serve her seven questions, followed Shneiderman's mantra and gave each one a keyboard equivalent. Both screens open with data visible: both hotels, the whole period and no country selected. Zoom, pan and drag were rejected because no chart shows more than 26 months or 12 countries.

#	Interaction	Course type	Mantra step
I1	Hotel filter	Manipulation: filter	Zoom and filter
I2	Year filter	Manipulation: filter	Zoom and filter
I3	Filters kept between screens	Exploration: navigation	Overview first
I4	Reset filters	Manipulation: filter	Zoom and filter
I5	Linked month highlight	Problem solving: linked views	Details on demand
I6	Tooltips	Problem solving: details on demand	Details on demand
I7	Select a country	Problem solving: linked views	Details on demand
I8	Chart / Map switch	Exploration: change of view	Overview first
I9	Sort toggle	Manipulation: order	Zoom and filter
I10	Play the year	Exploration: animation	Overview first
I11	Table view	Problem solving: details on demand	Details on demand
I12	Night mode	Manipulation: customisation	(course H18)

Interactions between charts

The hotel and year filters (I1, I2) act on all KPIs and charts of a screen, and the link to the other screen carries them (I3). The year labels name the partial years, "2015 (Jul–Dec)" and "2017 (Jan–Aug)". A reset button (I4) is always visible. On the main screen, pointing at a month or reaching it with the arrow keys marks it in the three charts (I5): a rule in the two line charts and an outline in the heatmap, with its values in the filter panel. On the detail screen, clicking a dot, bubble, box or country, or using the list or Previous and Next, selects a country in every chart and KPI (I7); a second click or the "No country" button clears it. A country outside the top 10 is added at the top of the country charts. The selection is shown by a band, a bold label, an outline and stripes, never by colour alone.

Interactions inside each chart

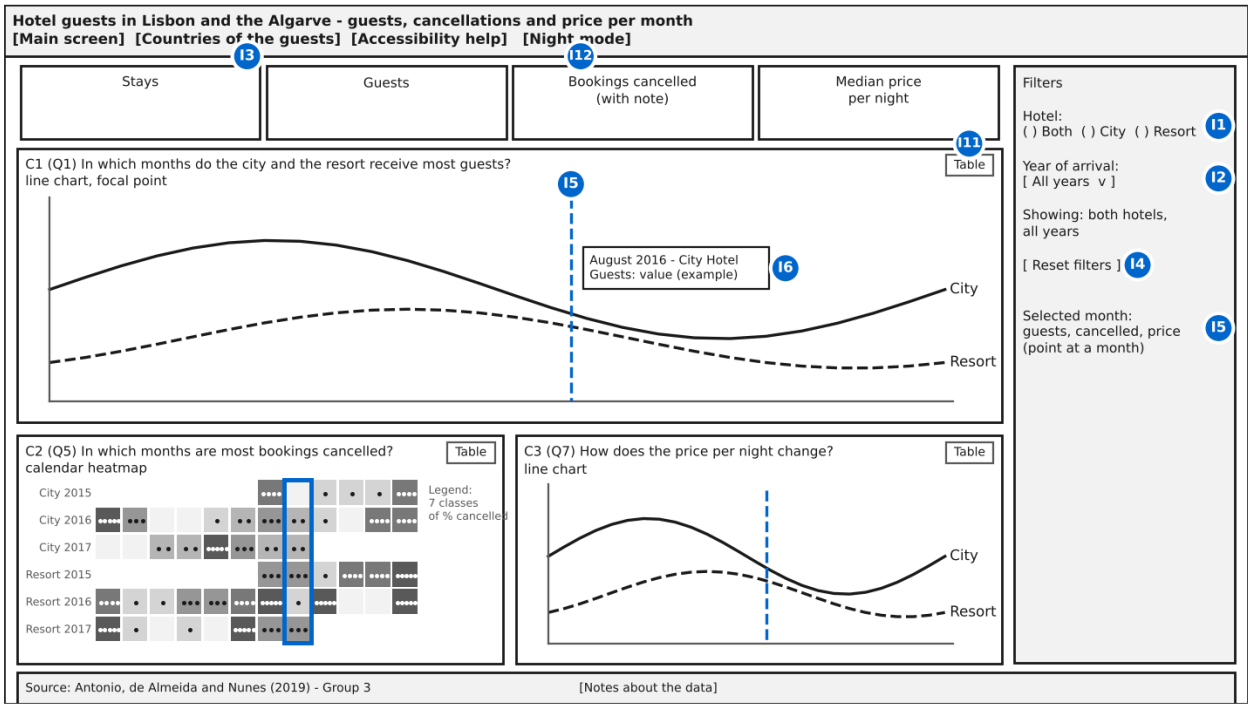
Tooltips (I6) give exact values with units, also for a mark reached with the keyboard. The sort toggle (I9) orders countries by guests or alphabetically. The Chart / Map switch (I8) shows the map in the dumbbell's box, keeping three charts visible. Play (I10) moves the bubbles from January to December; it never starts by itself and has Pause and a month slider (WCAG 2.2.2). Each chart has a Table button (I11), and night mode (I12) darkens both screens.

Feasibility

Every chart feature was built in a small Vega-Altair test and compiled to SVG; the switch, tables, reset and links need only HTML. Because each chart is embedded separately, with its own alternative text, selections are passed between charts with a few lines of JavaScript, built and tested in Part 6. Figures 1 and 2 show where each interaction appears.

Figures

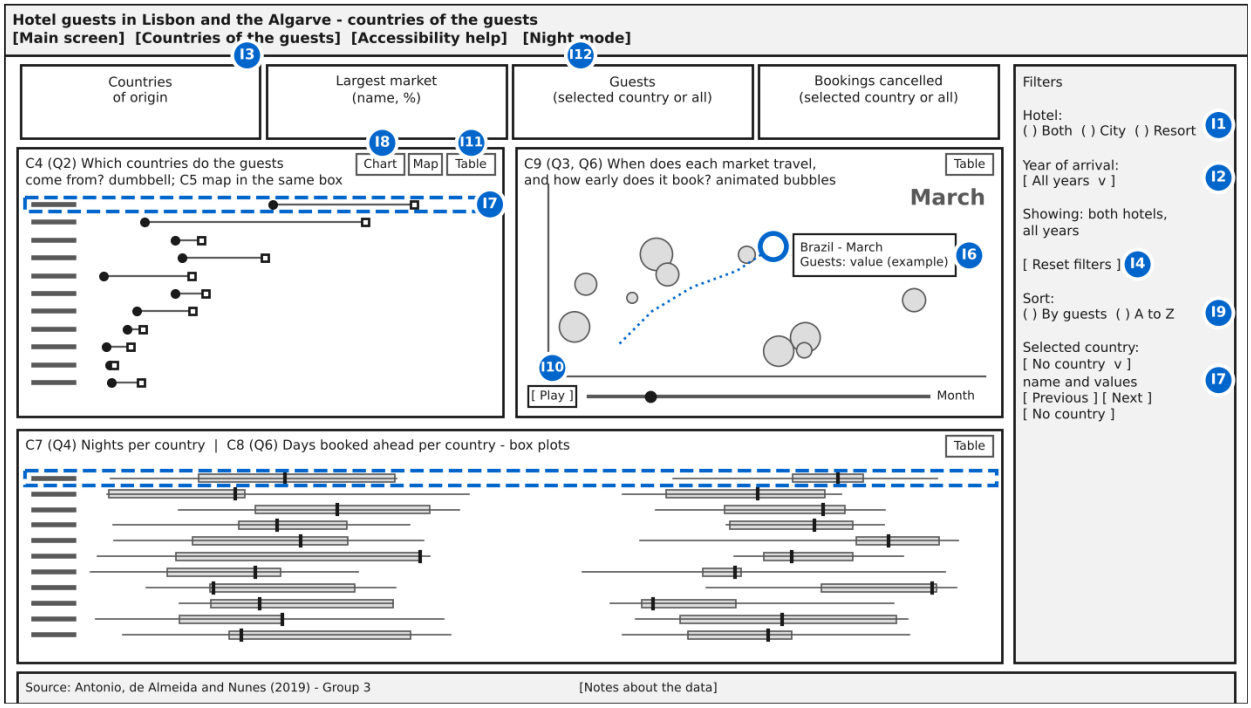
Prototype with interactions: main screen



Interactions: I1 Hotel filter I2 Year filter I3 Filters kept between screens I4 Reset filters I5 Linked month highlight I6 Tooltips I11 Table view I12 Night mode

Figure 1. Main screen with interactions I1 to I6, I11 and I12. Source: our design; code: prototype_interactions.ipynb.

Prototype with interactions: detail screen



Interactions: I1 Hotel filter I2 Year filter I3 Filters kept between screens I4 Reset filters I6 Tooltips I7 Select a country I8 Chart / Map switch I9 Sort toggle I10 Play the year I11 Table view I12 Night mode

Figure 2. Detail screen with interactions I1 to I4 and I6 to I12. Source: as Figure 1.